EFY TESTED Ashwini K. Sinha

Part 2

IMPLANTABLE CHIPS: Storing Data Inside Body

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In the December 2023 issue of EFY, we explored the implantable chip-based data tracking system for animals. Now, let's proceed to design a device that allows us to securely store sensitive data and passwords inside the body through the process of storing data within a tiny chip. It can be implanted in a way that only you can access your sensitive information. Many individuals choose to implant chips containing data they wish to keep private. This safeguards against loss or leaks, making implantable chips an increasingly popular choice.

There are two main approaches:

Authentication chip with data unlocking. You can implant a chip that serves as an authentication system, unlocking digital data stored in the cloud or on a laptop. Access to the data is granted only when scanning the chip inside the body.

Memory-embedded chip. Or you can implant a chip with a small memory that stores written data within itself. This data can be read and accessed by scanning the chip.

Implantable chip with data inside

This method ensures maximum security, keeping important data within your body in an implanted chip, beyond the reach of the outside world. We use an implantable chip with both read and write capabilities. Several options are available in the market, such as:

xEM RFID implant. An RFID implant by Dangerous Things, operating at 13.56MHz with both read and write capabilities.

xCARD RFID implant. Another offering by Dangerous Things, a 13.56MHz RFID implant suitable for various applications, including data storage.

NFC (near field communication) implant. Operating at 13.56MHz, NFC implants, while not precisely RFID, can store and retrieve data.

Biohacking implant. Various biohacking communities and companies explore RFID and NFC devices with customisable features, including data storage.

Choose any of these implants to write your data securely.

All the materials (components) are listed in the bill of materials table.

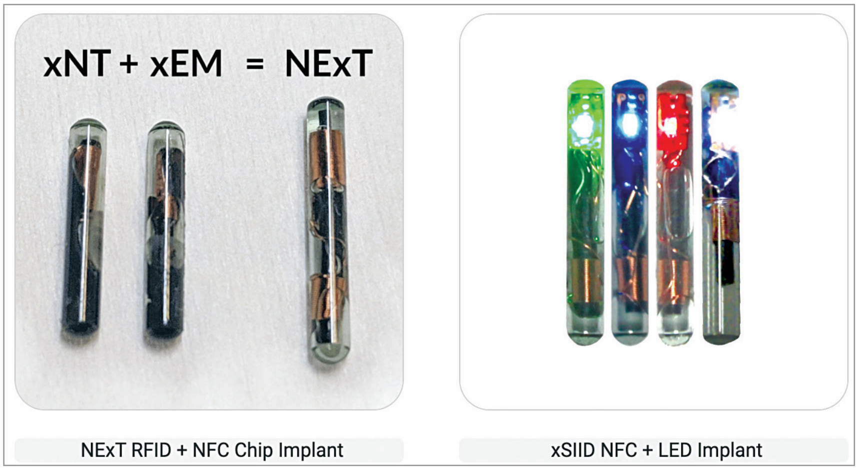

Fig. 1: Images of some implantable chips (Source: https://dangerousthings.com)

BILL OF MATERIALS

Components	Quantity	Description
Implantable RFID capsule	1	13.56MHz read and write RFID capsule
13.56MHz RFID reader/writer	1	Serial reader/writer module for the chip
Raspberry Pi Zero / 4 (optional)	1	SBC

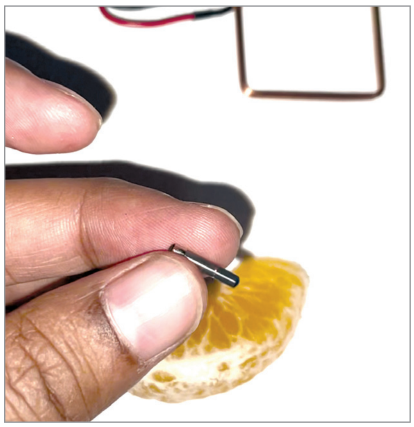

Fig. 2: Implanting the chip inside an orange segment for testing

In the first process, we write